

ContractGuard

Catching Construction Billing Leaks Before Payment Goes Out

Kaya AI IIT India Hackathon 2026

Track 3: Procurement

Mohar · Harshita · Dharmik

Agenda

- The Problem
- The Solution
- Live Demo Results
- Why ContractGuard Wins

The Problem

Construction billing in India runs on BOQ-based RA bills - reconciled by hand, from records scattered across WhatsApp.

Every RA bill must be checked against the BOQ AND every certified bill before it. Manual review can't keep up.

Three leaks slip through every time:

- Rate drift vs the BOQ
- Quantity double-billed across bills
- Wrong GST rate vs HSN code

By the time it's caught, the payment is already gone.

The Solution

ContractGuard reconciles a project's full BOQ and RA bill history automatically — before payment goes out, not after.

Every flag it raises includes:

- rupee impact
- confidence score
- exact citation
- a ready-to-send vendor follow-up

How It Works

1. Ingest BOQ + full RA Bill history

2. Run 3 deterministic checks

Rate drift , Cumulative cross-bill check , GST vs HSN

3. Every flag: WHERE → WHAT → RUPEE IMPACT → CONFIDENCE → CITATION

4. One click: draft the vendor follow-up email

Demo Project

Shree Balaji Residency, Block B

Contractor: M/s Verma Constructions

1 BOQ (15 line items) + RA Bills 1, 2, 3

What We Caught

E1 · RA Bill 1, Steel

GST charged 28% vs correct HSN rate 18% ₹1,17,000

E2 · RA Bill 3, Excavation

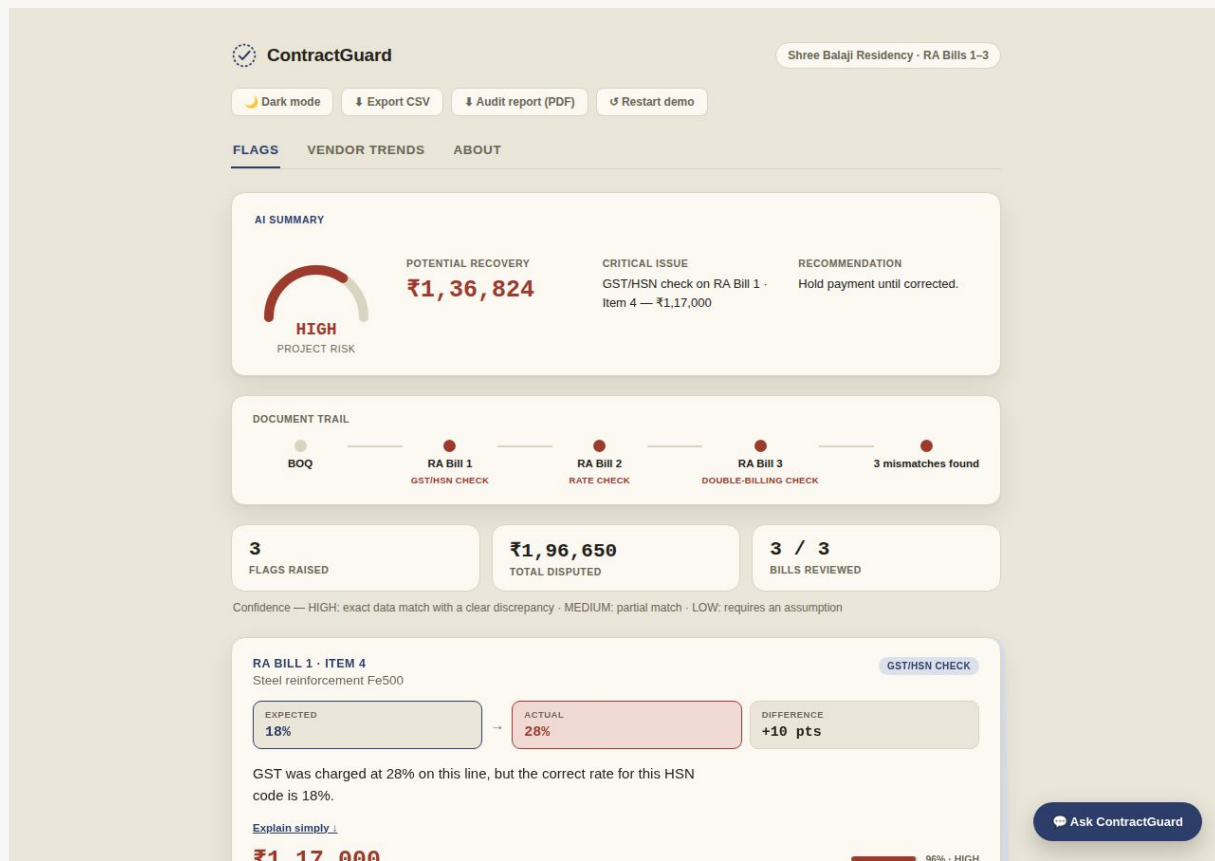
250 m³ double-billed across bills ₹53,100

E3 · RA Bill 2, Brickwork

₹995/m² charged vs BOQ rate ₹950/m² ₹26,550

TOTAL LEAKAGE CAUGHT: ₹1,96,650

See It In Action



Every flag:

- WHERE
- WHAT
- RUPEE IMPACT
- CONFIDENCE
- CITATION

Plus a one-click Draft Vendor Email action.

Why ContractGuard

Problem Relevance

Silent, recurring rupee leakage on every project — solved at the source.

Technical Depth

Deterministic reconciliation across full bill history — not a passive tracker.

Impact & Scale

Scales with vendor count and project volume, with zero added manual effort.

Originality

Most teams build dashboards that watch. ContractGuard catches, explains, and acts — before the money leaves.

Team

Mohar Biswas— Document Intelligence

Harshita Manav— Mismatch Detection + Deck

Dharmik Patel— Interface, Agentic Layer & Demo